

AWS Intern – Hands-On Task

Highly-Available Web Application

Overview

Build a production-style, highly-available web application on AWS using only the AWS Management Console (CFT/Terraform optional). The goal is to demonstrate end-to-end hands-on skills across VPC networking, compute, load balancing, auto scaling, and security best practices.

This task is expected to take approximately 4–6 hours and should be completed as a take-home assignment.

Objectives

- Design and provision a multi-AZ VPC with proper public/private subnet separation.
- Deploy an Application Load Balancer (ALB) that distributes traffic across multiple EC2 instances.
- Configure an Auto Scaling Group (ASG) to maintain high availability and self-healing capability.
- Apply least-privilege security group rules.
- Demonstrate operational understanding by validating, documenting, and tearing down the setup.

Part 1 – Infrastructure Requirements

Provision the following components through the AWS Console:

1. Custom VPC spanning 2 Availability Zones.
2. 2 Public Subnets (one per Availability Zone).
3. 2 Private Subnets (one per Availability Zone).
4. 1 Internet Gateway attached to the VPC.
5. 1 NAT Gateway placed in a public subnet (to allow private EC2 instances outbound internet access for package installs and updates).
6. Route Tables configured so that: public subnets route traffic to the Internet Gateway, and private subnets route traffic to the NAT Gateway.
7. Security Groups following the least-privilege model.

Part 2 – Application Layer Requirements

Deploy the following:

8. Application Load Balancer (ALB) deployed in the two public subnets.
9. Auto Scaling Group with a minimum of 2 EC2 instances launched in the private subnets.
10. EC2 instances must run NGINX (or a Dockerized sample app) serving a page that displays the instance ID or Availability Zone. This makes load balancing visible to the evaluator.
11. ALB Target Group with health checks configured (e.g., HTTP GET / every 30 seconds).
12. Verify that the Auto Scaling Group automatically replaces any manually terminated instance.

Evaluation Criteria

Your submission will be evaluated against the following checks:

#	Check	What it validates
1	Visit the ALB DNS in a browser and refresh several times — different instance IDs / AZs should appear.	Load balancing actually works.
2	Manually terminate one EC2 instance — the ASG should launch a replacement within a few minutes.	Auto Scaling and self-healing.
3	Try to SSH directly into a private EC2 from your laptop — it should fail.	Private subnets are truly private.
4	Review the EC2 Security Group — it should reference the ALB Security Group.	Least-privilege networking.
5	Complete teardown of all resources after evaluation.	Cost awareness and operational discipline.

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